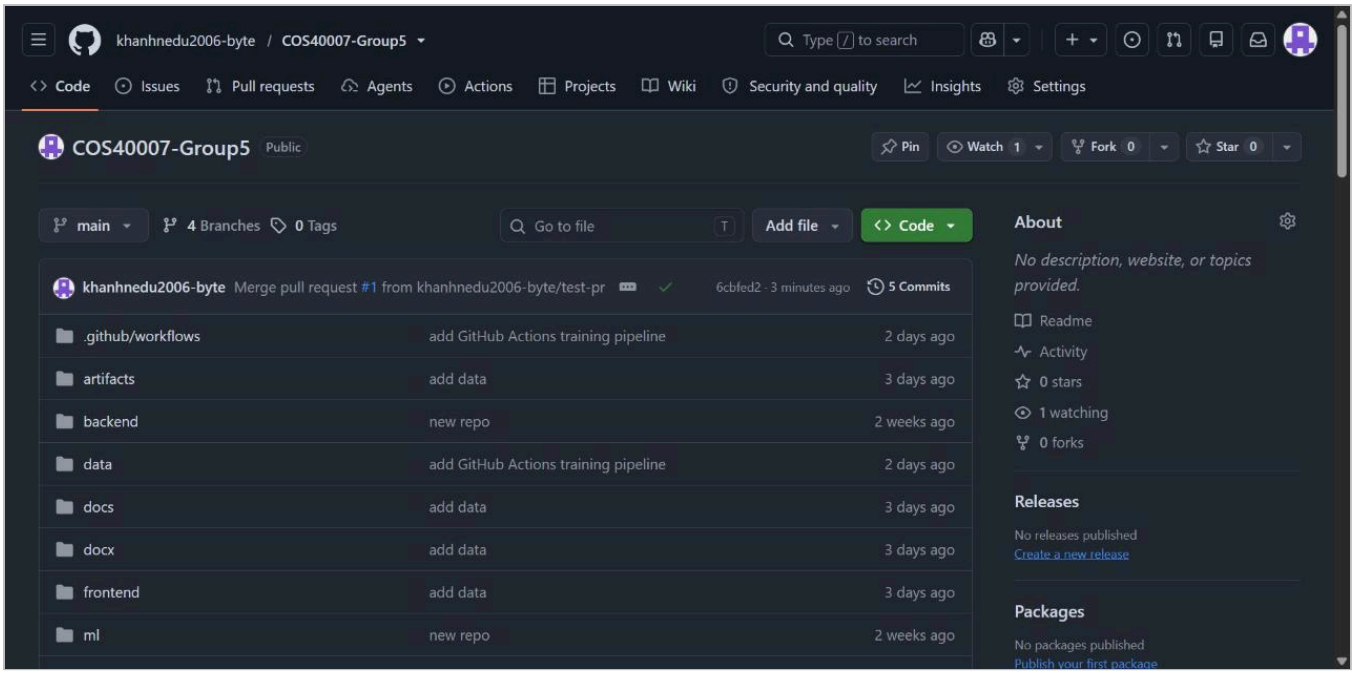
Group Task 2: GitHub Actions ML Pipeline

COS40007 Artificial Intelligence Engineering · Group 5 · Project: Mobile Internet Speed Prediction for Hanoi
Members: Nguyễn Nam Khánh, Hoàng Minh Đức, Phạm Quang Minh
Repository: https://github.com/khanhnedu2006-byte/COS40007-Group5

Repository & File Structure

The group repository **COS40007-Group5** contains `.github/workflows/train.yml`, `model.py` and `requirements.txt` in the required locations.



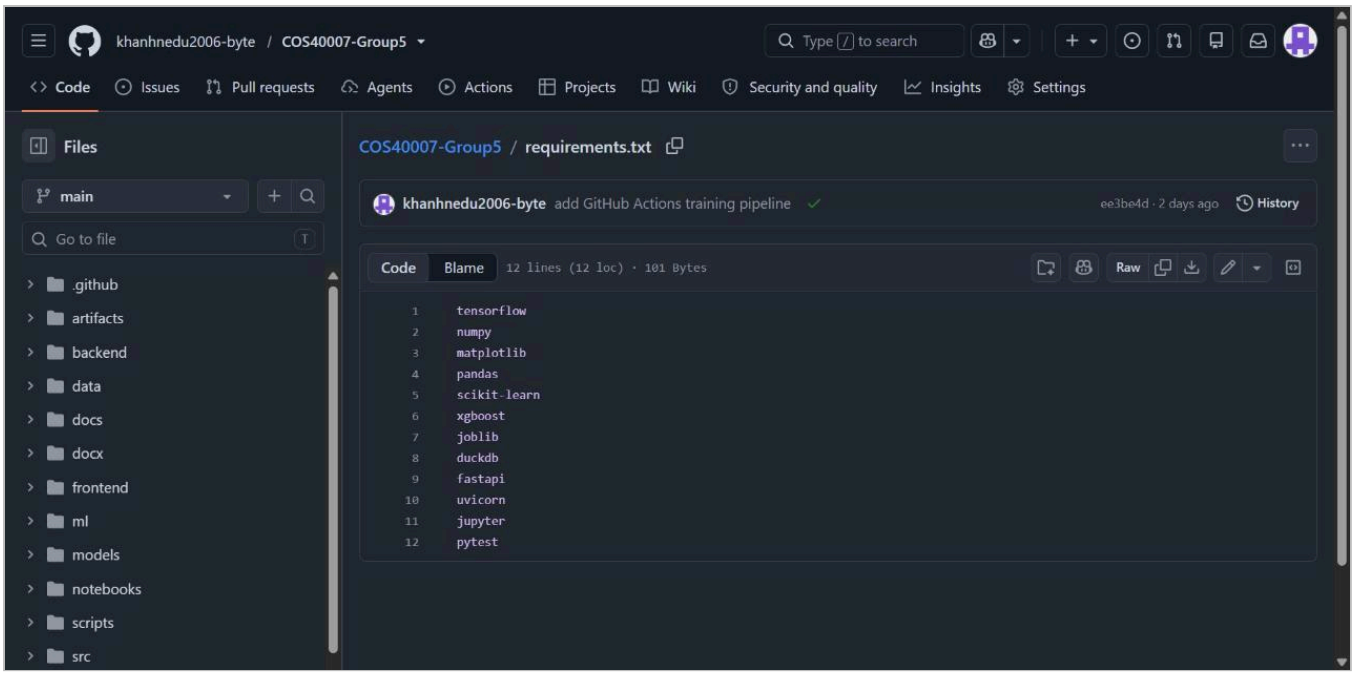
Repository home page (https://github.com/khanhnedu2006-byte/COS40007-Group5): folder `.github/workflows`



Same page scrolled down: `model.py` and `requirements.txt` in the repository root

requirements.txt

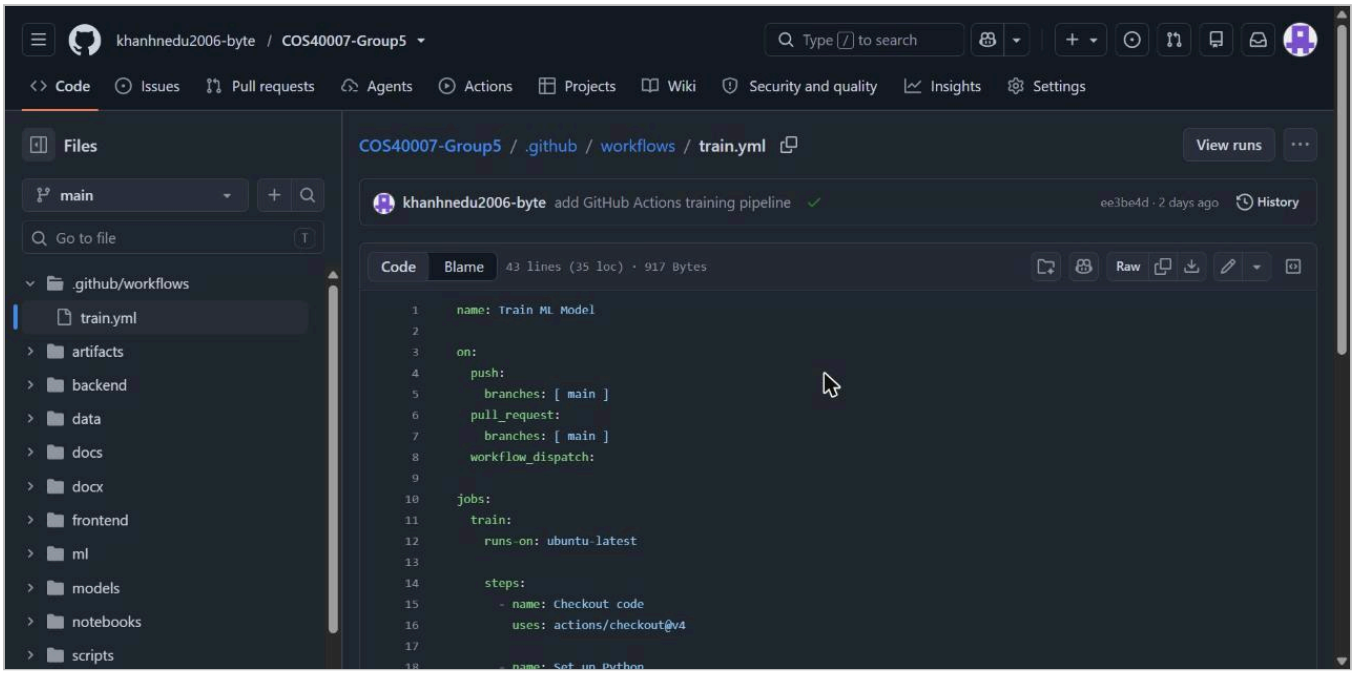
Includes `tensorflow`, `numpy` and `matplotlib`, plus the libraries used by the rest of our project.



https://github.com/khanhnedu2006-byte/COS40007-Group5/blob/main/requirements.txt

Workflow Triggers (on:)

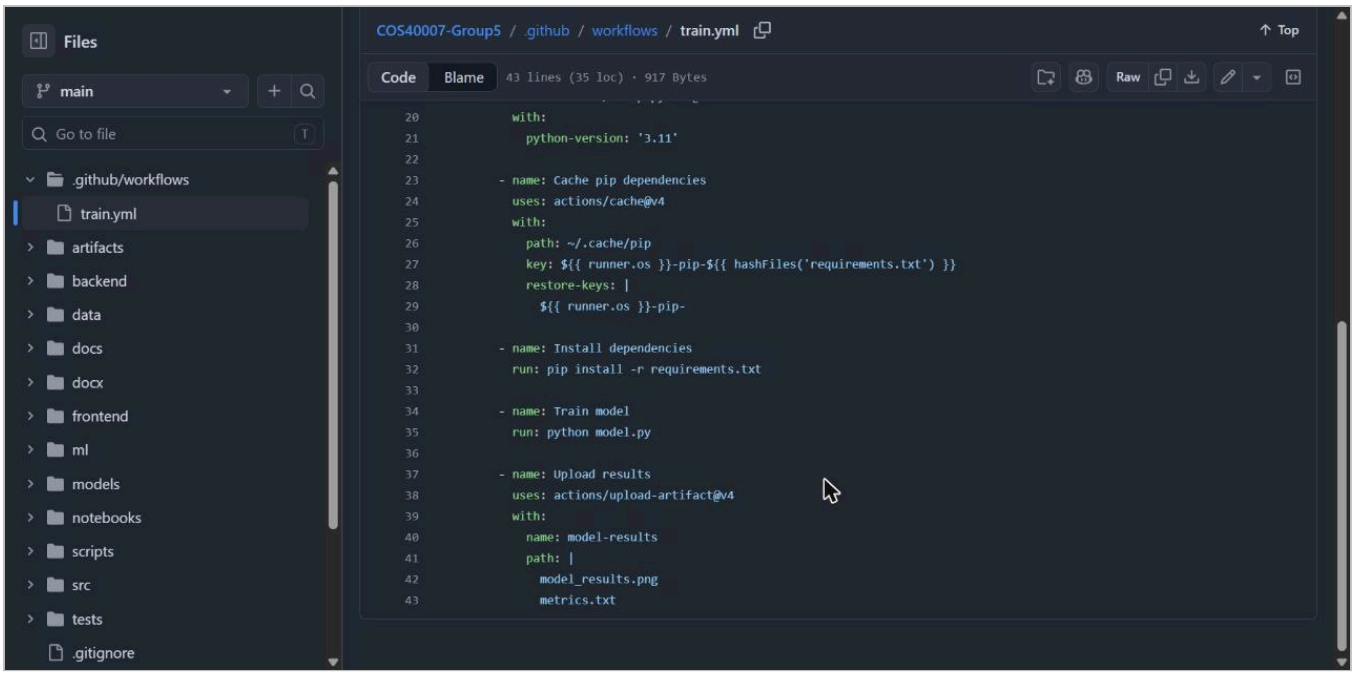
The workflow runs on `push` to main, `pull_request` to main and manual `workflow_dispatch` (lines 3–8).



https://github.com/khanhnedu2006-byte/COS40007-Group5/blob/main/.github/workflows/train.yml (top)

Pipeline Job Definition

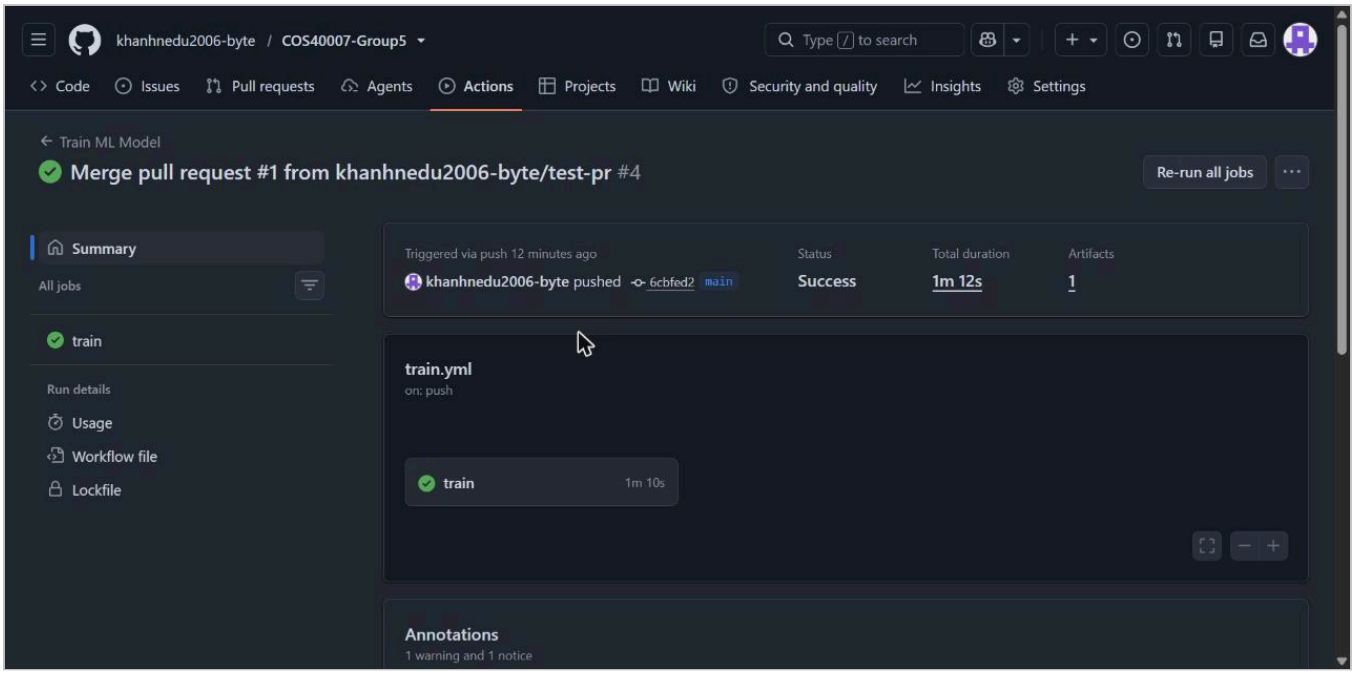
Job `train` runs on `ubuntu-latest` with the steps checkout → setup-python (3.11) → cache pip → install requirements → run `model.py` → upload artefacts (`actions/upload-artifact@v4`).



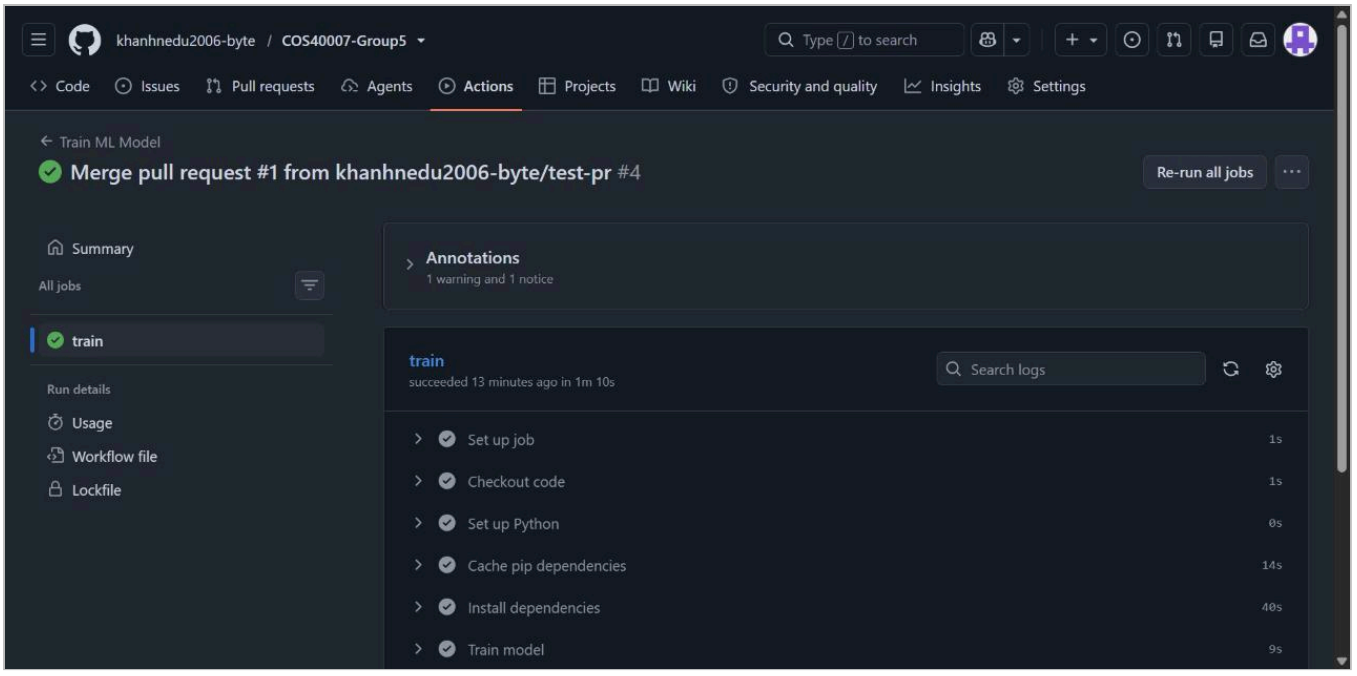
train.yml (bottom): cache, install, train and upload steps

Successful Pipeline Execution

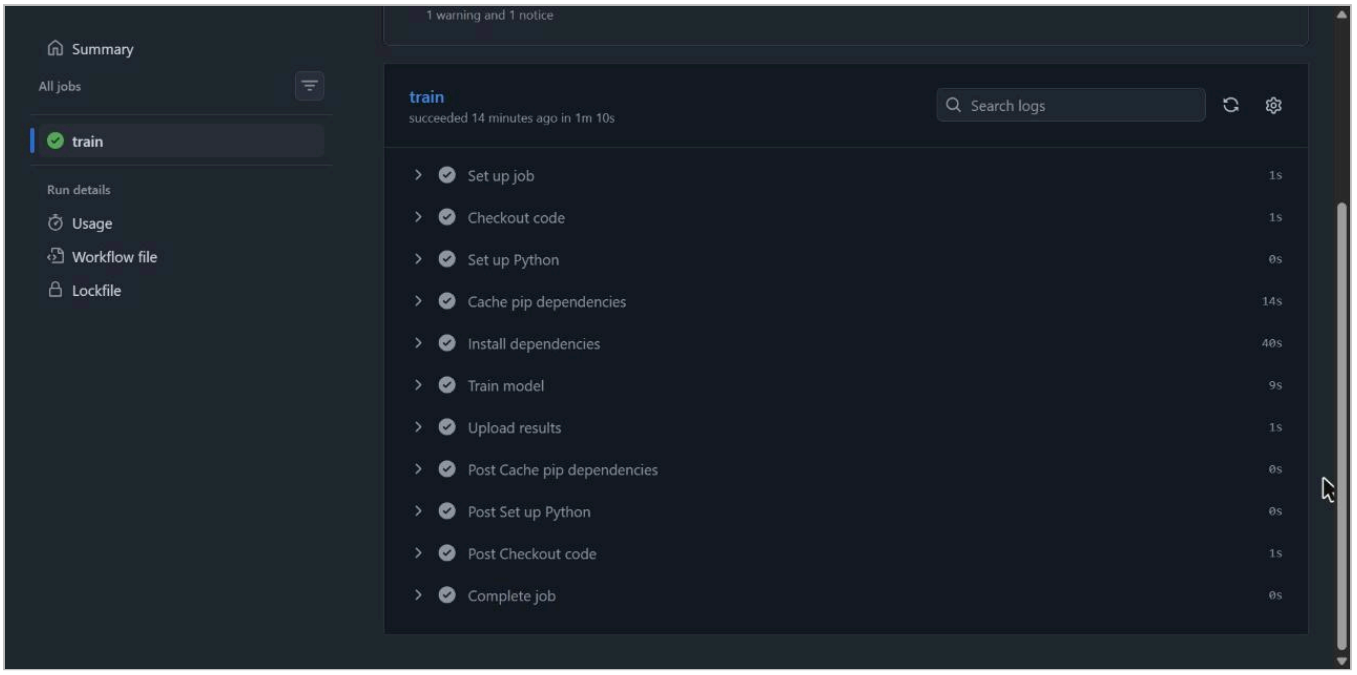
Run #4 (push to main after merging the pull request) finished with status **Success**; every step of the `train` job has a green check. Our `model.py` trains a TensorFlow neural network that predicts mobile download speed in Hanoi from location and cell-tower features.



Run summary: Status Success, 1 artifact (https://github.com/khanhnedu2006-byte/COS40007-Group5/actions/runs/41306240036)



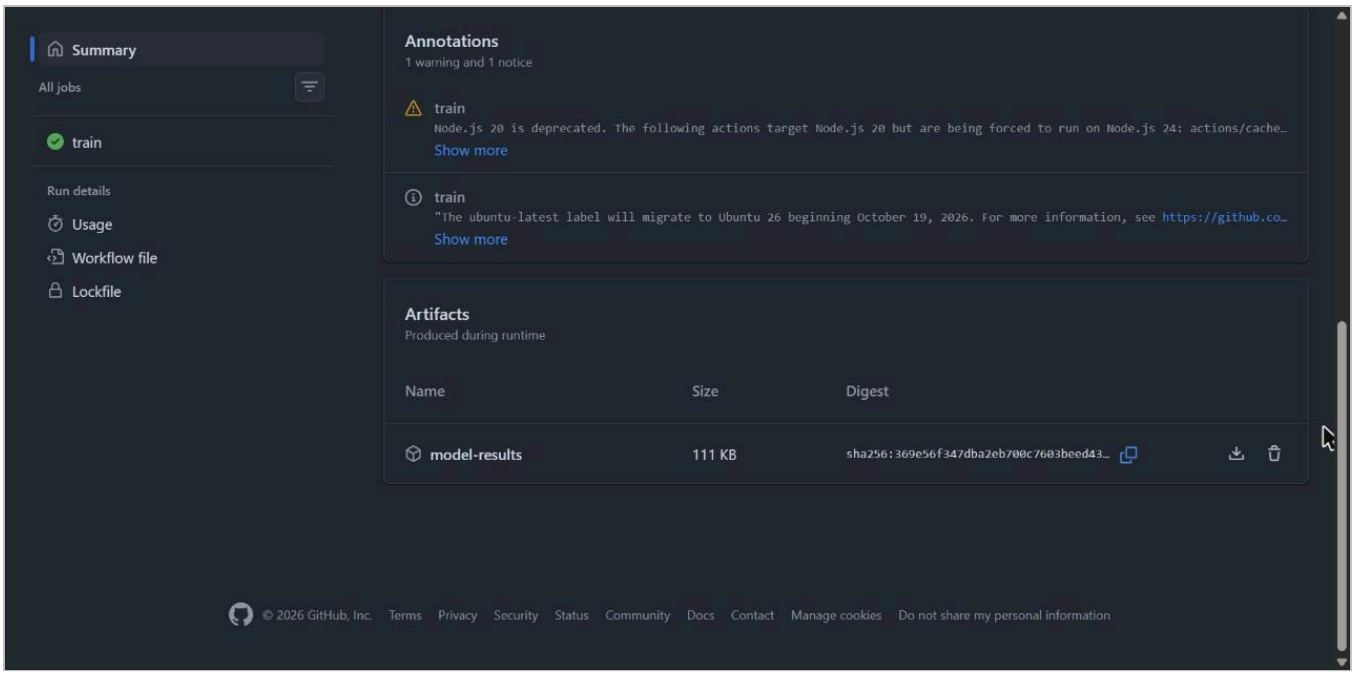
Job `train` succeeded



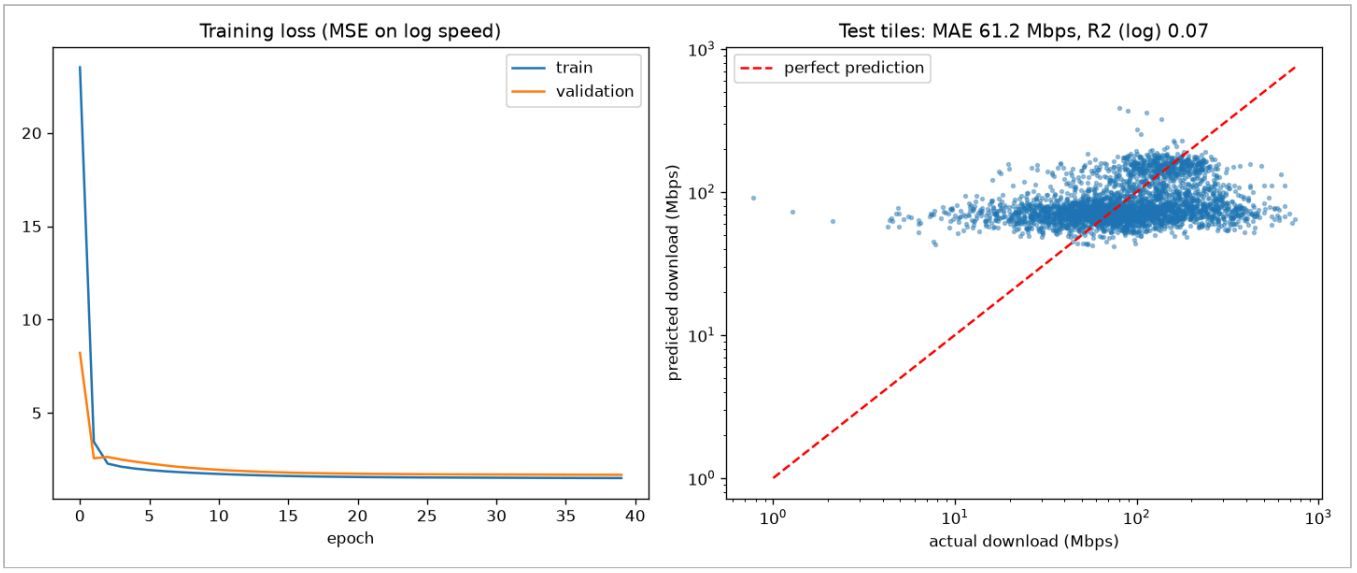
All steps green: Checkout code, Set up Python, Cache pip dependencies, Install dependencies, Train model, Upload results

Artefact Generation & Upload

`model.py` generates `model_results.png` and `metrics.txt`; the workflow uploads both as the artefact **model-results**.



Artifacts section of the run: `model-results` (111 KB)



Downloaded `model_results.png`: training loss and predicted vs actual download speed on test tiles

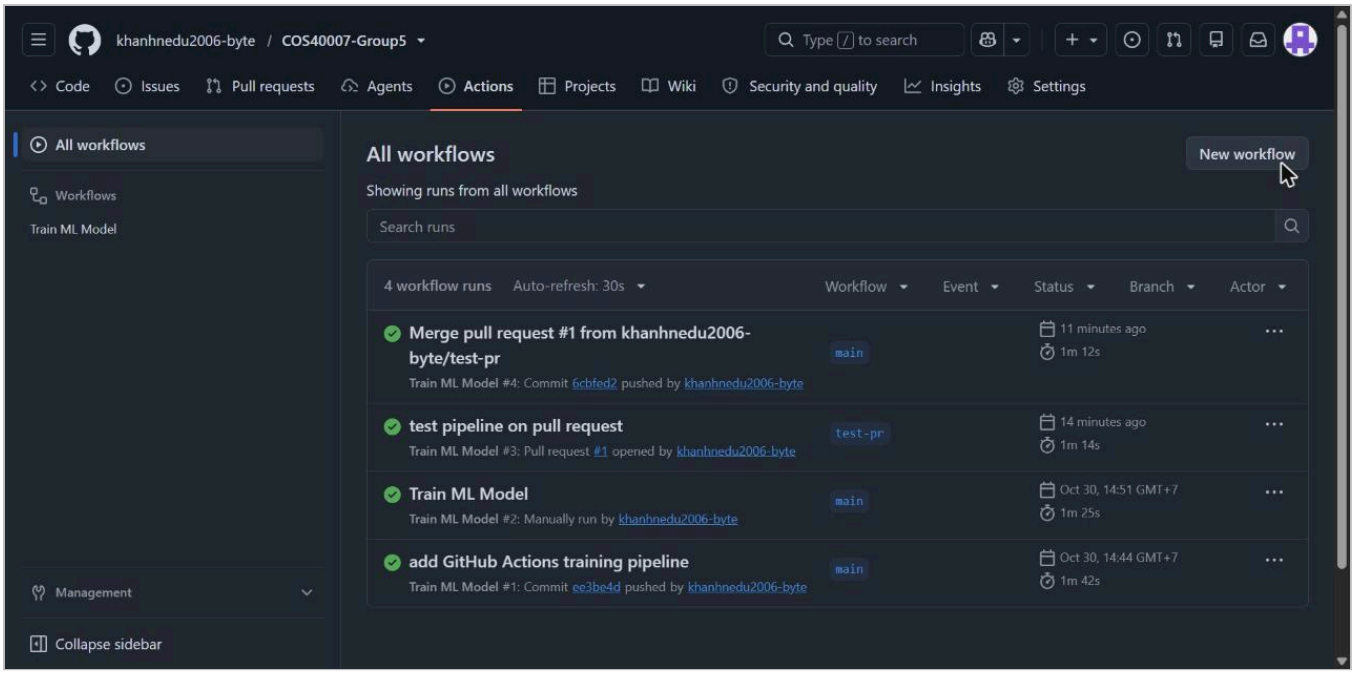
```
Hanoi mobile download speed - neural network (TensorFlow)
Data: 30,076 tile-quarter rows, 10,441 tiles, quarters 2024-4 to 2026-1
Split: spatial blocks (2 km), train 23,427 rows, test 3,211 rows with >= 3 tests
Features: latitude, longitude, dist_nearest_tower_m, towers_500m, towers_1km,
quarter_index
Epochs trained: 40

MAE  (Mbps): 61.18
RMSE (Mbps): 97.41
R2 (Mbps)      : -0.105
R2 (log speed) : 0.066
Quality level accuracy: 57.3%
Baseline MAE (median of training set): 67.38
Improvement over baseline: 9.2%
```

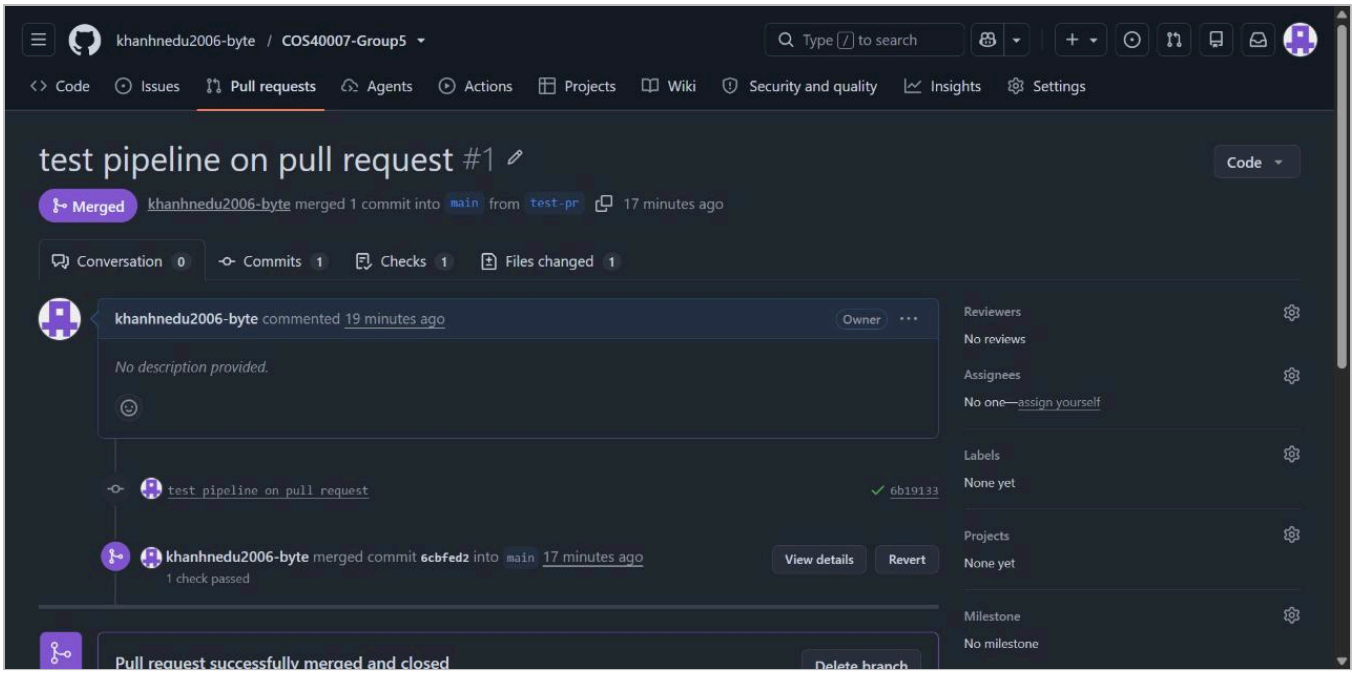
Downloaded `metrics.txt`

Demonstration of Functionality

The pipeline was triggered in all three ways and the artefact was downloaded from the Actions UI: run #1 by **git push**, run #2 **manually** (Run workflow button), run #3 by a **pull request** (PR #1, check passed), and run #4 by the push that merged the pull request.



https://github.com/khanhnedu2006-byte/COS40007-Group5/actions: four successful runs (push, manual, pull request)



https://github.com/khanhnedu2006-byte/COS40007-Group5/pull/1: pull request merged, 1 check passed

GitHub Link

https://github.com/khanhnedu2006-byte/COS40007-Group5